WEG W22 75kW Three-Phase Induction Motor — Service Manual

Document number: MAN-MTR-002   Revision: Rev. B
Published: 2023-11-08T00:00:00Z

EQUIPMENT
Tag: EQ-MTR-W22-75-04   Plant: North Refinery Unit 4
Model: W22 IE3 75kW 4P   Category: AC Induction Motor
Serial: WEG-22-99041-A
Manufacturer: WEG (Brazil)

SAFETY HAZARDS
- Electrical shock (460V)
- Arc flash
- Suspended load during lift

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Section 3 — Bearing Replacement

Procedure ID: PRC-MTR-002-A
Category: Corrective Maintenance   Severity: High
Performed by: T-1042 Priya Subbu (Electrical)

OVERVIEW
This procedure covers replace bearings (drive and non-drive end) on equipment EQ-MTR-W22-75-04 (W22 IE3 75kW 4P). Follow the safety hazards listed on page 1 before starting. Lock out and tag out upstream isolation before disassembly.

PARTS REQUIRED
- BRG-7308BEP — Angular Contact Bearing 40mm (Bearings)
- BRG-6206-2RS — Sealed Deep-Groove Ball Bearing 30mm (Bearings)
- SEAL-LIP-32 — Lip Seal 32mm Nitrile (Seals)

STEPS
1. Confirm isolation per plant LOTO procedure. Verify zero energy state.
2. Stage the parts listed above plus a calibrated torque wrench and clean rags.
3. Disassemble per Section 3 of this manual.
4. Inspect mating surfaces for wear, scoring, or corrosion. Photograph any anomalies.
5. Install replacement parts in the order listed. Torque fasteners per Table 2.
6. Reassemble and reinstate auxiliaries (lubrication, cooling, instrumentation).
7. Remove isolation and perform a no-load function test.
8. Record completion against work order in CMMS.

ACCEPTANCE CRITERIA
- No abnormal vibration or noise during no-load run.
- All fasteners torqued and witness-marked.
- No leaks at static or dynamic seals after a 5-minute run.

Section 4 — Electrical Testing

Procedure ID: PRC-MTR-002-B
Category: Predictive Maintenance   Severity: Low
Performed by: T-1042 Priya Subbu (Electrical)

OVERVIEW
This procedure covers megger insulation resistance test on equipment EQ-MTR-W22-75-04 (W22 IE3 75kW 4P). Follow the safety hazards listed on page 1 before starting. Lock out and tag out upstream isolation before disassembly.

PARTS REQUIRED
- BRSH-CRBN-12 — Carbon Motor Brush 12mm Pair (Electrical)

STEPS
1. Confirm isolation per plant LOTO procedure. Verify zero energy state.
2. Stage the parts listed above plus a calibrated torque wrench and clean rags.
3. Disassemble per Section 4 of this manual.
4. Inspect mating surfaces for wear, scoring, or corrosion. Photograph any anomalies.
5. Install replacement parts in the order listed. Torque fasteners per Table 2.
6. Reassemble and reinstate auxiliaries (lubrication, cooling, instrumentation).
7. Remove isolation and perform a no-load function test.
8. Record completion against work order in CMMS.

ACCEPTANCE CRITERIA
- No abnormal vibration or noise during no-load run.
- All fasteners torqued and witness-marked.
- No leaks at static or dynamic seals after a 5-minute run.

Section 5 — Alignment Procedure

Procedure ID: PRC-MTR-002-C
Category: Reliability Inspection   Severity: Medium
Performed by: T-1208 Aisha Kone (Reliability Engineering)

OVERVIEW
This procedure covers coupling alignment to driven pump on equipment EQ-MTR-W22-75-04 (W22 IE3 75kW 4P). Follow the safety hazards listed on page 1 before starting. Lock out and tag out upstream isolation before disassembly.

PARTS REQUIRED
- CPLG-FLEX-32 — Flexible Jaw Coupling 32mm (Couplings)
- BLT-M8-FLG — Flanged Bolt M8 x 30 Stainless (Fasteners)

STEPS
1. Confirm isolation per plant LOTO procedure. Verify zero energy state.
2. Stage the parts listed above plus a calibrated torque wrench and clean rags.
3. Disassemble per Section 5 of this manual.
4. Inspect mating surfaces for wear, scoring, or corrosion. Photograph any anomalies.
5. Install replacement parts in the order listed. Torque fasteners per Table 2.
6. Reassemble and reinstate auxiliaries (lubrication, cooling, instrumentation).
7. Remove isolation and perform a no-load function test.
8. Record completion against work order in CMMS.

ACCEPTANCE CRITERIA
- No abnormal vibration or noise during no-load run.
- All fasteners torqued and witness-marked.
- No leaks at static or dynamic seals after a 5-minute run.